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Hand image segmentation using color and RCE neural network

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Abstract

This paper presents a color segmentation method based on RCE neural network for hand image segmentation in the gesture-based human–service robot interaction system. The study on skin color distributions in different color spaces indicates that skin colors cluster in a small region in a color space. The RCE neural network characterizes the skin color distribution region using skin color prototypes together with their spherical influence fields during training stage, and identifies the skin regions in the color image during running stage. Experimental results have demonstrated the effectiveness of this method for the segmentation of various hand images as well as general color images with complex backgrounds. © 2001 Elsevier Science B.V. All rights reserved.

Keywords: Hand image segmentation; RCE neural network; Skin color distribution; Color segmentation

1. Introduction

With the massive influx of computers in society and the increasing importance of service sectors in many of industrialized nations, the market for robots in conventional applications of manufacturing automation is reaching saturation, and the research on robotics is rapidly proliferating in the field of service industries [1,2]. Service robots operate in the dynamic and unstructured environment and interact with people who are not necessarily skilled in handling robots [3]. Friendly and cooperative interface is thus critical for the development of service robots [4,5]. Vision-based hand gesture recognition holds the promise of making human–robot interaction more natural and efficient [6].

Hand image segmentation separates the hand image from the background. It is the first important step in any hand gesture recognition system, and all subsequent stages heavily rely on the quality of segmentation. Image segmentation has been regarded as one of the toughest problems in computer vision community. In the domain of human–service robot interaction, it is a more challenging problem because of the following special difficulties: the background is complex and dynamic; the lighting condition is variable; the shape of the human hand is deformable; the implementation is required to be executed in real time and the system is expected to be user and device independent [6].

Two types of cues, color cues and motion cues, are often applied for hand image segmentation [7]. Motion cues are used in conjunction with some assumptions [8,9]. For example, the gesturer is stationary with respect to the background that is also stationary. Hence, the main component of motion in the visual image is the motion of the hand of the gesturer. Such assumption restraints its application at the occasions when the background is not stationary. Color

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cues are applicable because of the characteristic color of the human skin [10,11]. Most color segmentation approaches depend on histogram matching [12] or employ simple look-up-table technique [13,14]. The disadvantage of these approaches is that the variability of the skin color under different lighting conditions frequently results in undetected skin regions or falsely detected non-skin regions. Some researchers alleviate this problem by using controlled imaging conditions such as light tables [15] and special backgrounds [16], or resorting uniquely colored gloves or marks on hands [17–19].

Kjeldsen and Kender [14] have used skin color to segment hand images for a gesture-based user interface. The segmentation was based on a histogram-like structure called a color predicate (CP). The system operated in office environments with typical office lighting conditions and relied on the assumption that skin coloration was relatively unique in the target environment. In the work of Triesch [6], the combination of motion and color cues was used to track the human hand for a gesture-based human–robot interaction system, and additional stereo cue was used to rule out targets which are not in the plane of fixation of the robot. Skin color detection of this system was based on a defined prototypical skin color point in the *HS* plane, which had the drawbacks of being prone to noise and sensitive to the change of the illumination. The Perseus system developed by Kahn [20] was used to recognize the pointing hand gesture. In the system, a variety of information about the image, such as intensity, edge, motion, disparity, and color, has been used to segment the hand image. The system was only used in a restricted indoor environment.

In this paper, we present a novel color segmentation approach for hand image segmentation. The study on the color distributions of the human skin in *RGB*, *HSI* and $L^*a^*b^*$ color spaces indicates that human skin colors distribute in a small region in a color space and have more difference in intensity than in color. The adaptive hand image segmentation algorithm is developed on the basis of Restricted Coulomb Energy (RCE) neural network. In this method, the color of the human skin is described by the skin color prototype and its spherical influence field. The skin color prototype is the point defined by the skin color in the color space, and derived from the training procedure of the RCE network. With the abundant skin color prototypes and their influence fields, the RCE network is capable to characterize the distribution region of skin colors accurately in the color space and identify the skin regions efficiently in the color image. This segmentation approach can not only segment various hand images but also general color images with complex backgrounds.

Based on this hand image segmentation algorithm, we have also developed a new method for accurate recognition of hand postures, which extracts topological features of the hand from the binary image of the segmented hand region, and recognizes hand postures on the basis of the analysis of these features. A gesture-based human–robot interface has subsequently been implemented, in which eight hand postures with distinctive features are used for gesture-based programming of a humanoid service robot. The details on the hand gesture recognition and robot programming are presented in another paper, and this paper focuses on the discussion of the hand image segmentation using color that is a crucial step toward any successful vision-based human–robot interaction.

We begin with the introduction of *RGB*, *HSI* and $L^*a^*b^*$ color spaces and discuss the skin color distributions in these color spaces in Section 3. Section 4 describes the architecture, training mechanisms and response modes of the RCE neural network for hand image segmentation. The algorithm of hand image segmentation using RCE network is given in Section 5. Section 6 presents experimental results of hand image segmentation, and Section 7 illustrates the segmentation of general color images. Finally, conclusions are given in Section 8.

2. Color spaces

Color segmentation techniques rely on not only the segmentation algorithms, but also the color spaces used. *RGB*, *HSI*, and $L^*a^*b^*$ are the most commonly used color spaces in computer vision, and have all been applied in many proposed hand image segmentation techniques. In order to find which color space is the most suitable for hand image segmentation, we explored our color segmentation approach in these three color spaces, respectively.

RGB color space is the standard color space used in every computer system as well as television, video, etc. Our images captured by the video camera are represented in *RGB* color space.

HSI color space is referred to perceptual color space. It separates the luminance component from chrominance information and corresponds closely to the way that human eyes perceive colors. *RGB* color space can be converted to *HSI* color space using the following formulae [21]:

$$H = \begin{cases} \theta & G \geqslant B, \\ 2\pi - \theta & G < B, \end{cases}$$

$$S = 1 - \frac{3\min(R, G, B)}{R + G + B},$$

$$I = \frac{1}{\sqrt{3}}(R + G + B), \tag{1}$$

where

$$\theta = ar\cos\left\{\frac{\frac{1}{2}[(R - G) + (R - B)]}{\sqrt{(R - G)^2 + (R - B)(G - B)}}\right\}. \tag{2}$$

$L^*a^*b^*$ color space is a uniform color space defined by the CIE (Commission International de l'Éclairage) in 1976. It maps equal Euclidean distance in the color space to equal perceived color difference. The transformation from *RGB* to $L^*a^*b^*$ color space is defined as follows [22]:

$$L^* = 116\left(\frac{Y}{Y_n}\right)^{1/3} - 16,$$

$$a^* = 500\left[\left(\frac{X}{X_n}\right)^{1/3} - \left(\frac{Y}{Y_n}\right)^{1/3}\right],$$

$$b^* = 200\left[\left(\frac{Y}{Y_n}\right)^{1/3} - \left(\frac{Z}{Z_n}\right)^{1/3}\right], \tag{3}$$

where X/X_n, Y/Y_n and $Z/Z_n > 0.008856$. X, Y and Z are tristimulus values of the specimen, and obtained from the values of R, G and B as follows:

$$X = 0.607R + 0.174G + 0.201B,$$

$$Y = 0.299R + 0.587G + 0.114B,$$

$$Z = 0.066G + 1.117B. \tag{4}$$

X_n, Y_n and Z_n are tristimulus values of a perfect reflecting diffuser, which are selected to be 250.410, 255.000, and 301.655, respectively.

3. Distribution of skin colors

A common belief is that different people have different skin colors, but some studies show that such a difference lies largely in intensity than color itself [23]. In this paper, we quantitatively investigate the skin color distributions of different human hands as well as the skin color distributions of the same human hand under different lighting conditions in *RGB*, *HSI* and $L^*a^*b^*$ color spaces.

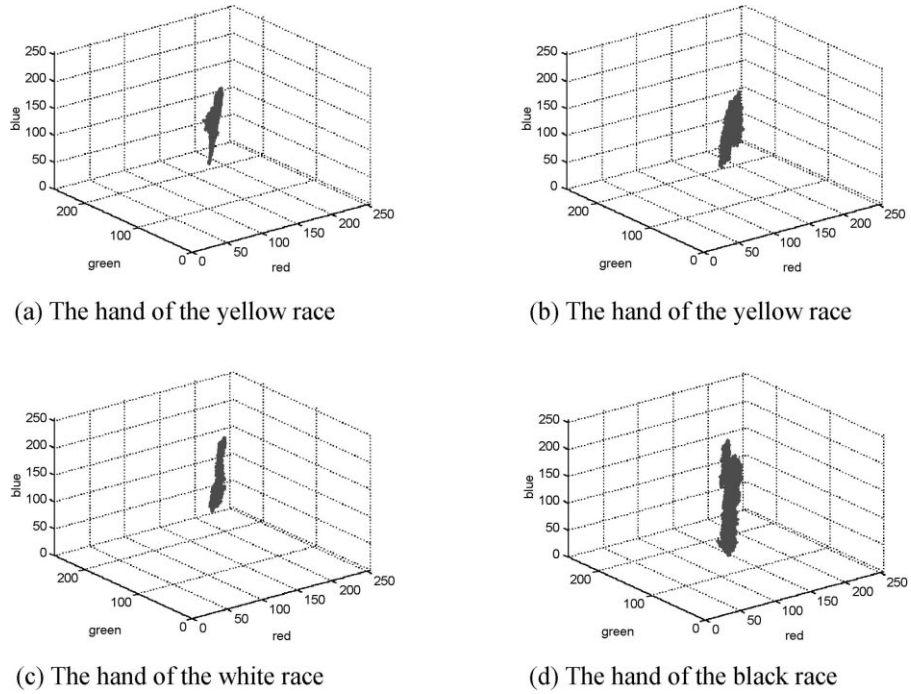
3.1. Skin color distribution of different human hands

Figs. 1–3 show the skin color distributions of different human hands in *RGB*, *HSI* and $L^*a^*b^*$ color spaces, respectively. (a)–(d) represent four typical hand images taken from people of different races and genders. The quantitative investigation indicates that skin color distributions of different human hands in *RGB*, *HSI*, and $L^*a^*b^*$ color spaces have the following properties:

1. Skin colors distribute in a small region in a color space.
2. Skin colors do not fall into a random region, but form clusters at specific points.
3. Skin color clusters are more concentrative in *HSI* and $L^*a^*b^*$ color spaces than in *RGB* color space.
4. Skin colors have more difference in intensity than in color.

3.2. Skin color distribution under different lighting conditions

Figs. 4–6 show the skin color distributions of the same human hand under different lighting conditions in *RGB*, *HSI* and $L^*a^*b^*$ color spaces, respectively. From (a) to (d), the light becomes darker and darker. It can be observed that skin color distributions under different lighting conditions have the following properties along with the change of lighting conditions:



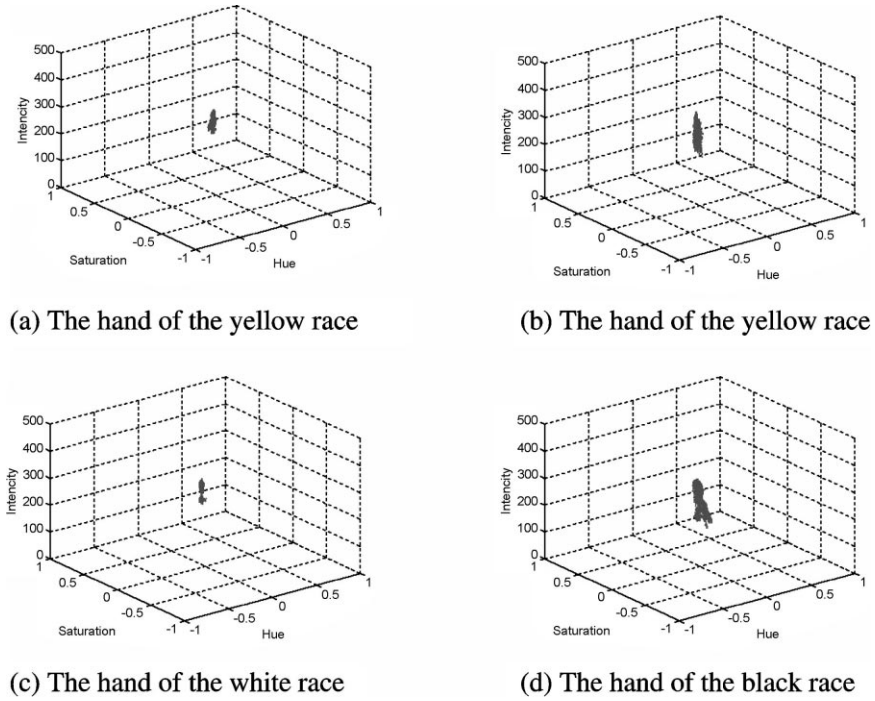
(a) The hand of the yellow race

(b) The hand of the yellow race

(c) The hand of the white race

(d) The hand of the black race

Fig. 1. Skin color distributions of different human hands in *RGB* color space.



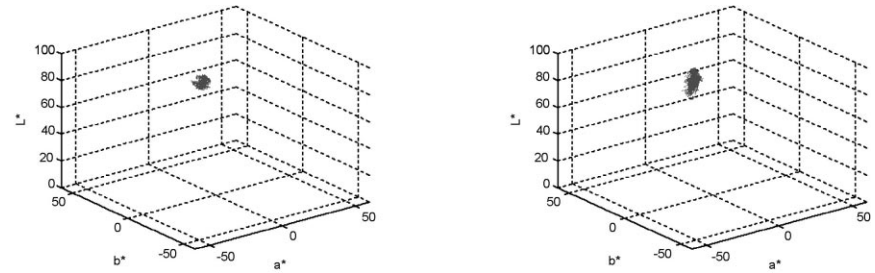
(a) The hand of the yellow race

(b) The hand of the yellow race

(c) The hand of the white race

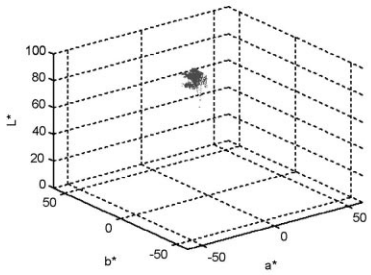
(d) The hand of the black race

Fig. 2. Skin color distributions of different human hands in *HSI* color space.

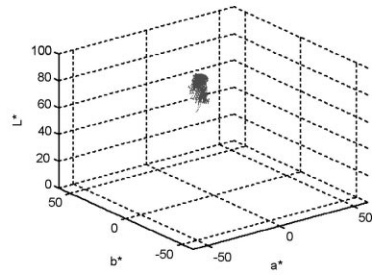


(a) The hand of the yellow race

(b) The hand of the yellow race

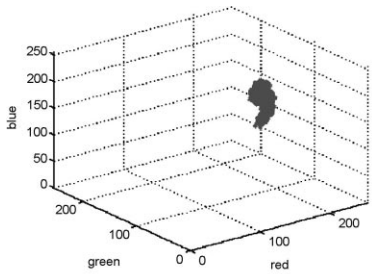


(c) The hand of the white race

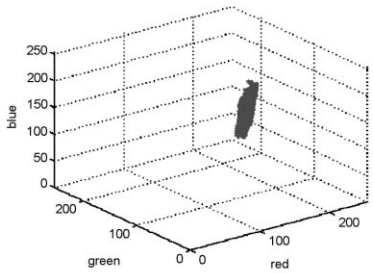


(d) The hand of the black race

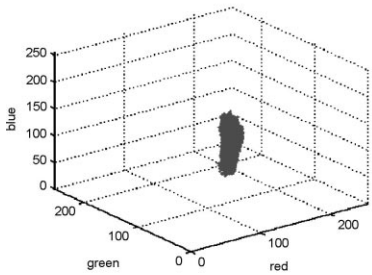
Fig. 3. Skin color distributions of different human hands in $L^*a^*b^*$ color space.



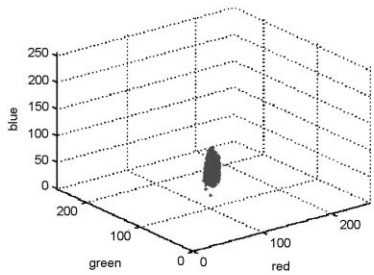
(a) In the brighter light



(b) In the bright light

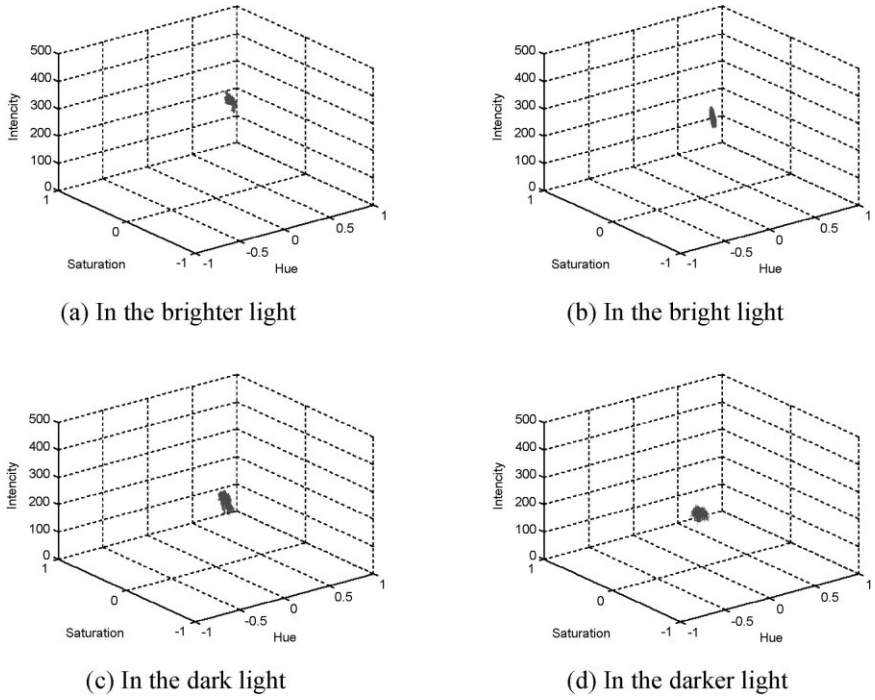


(c) In the dark light



(d) In the darker light

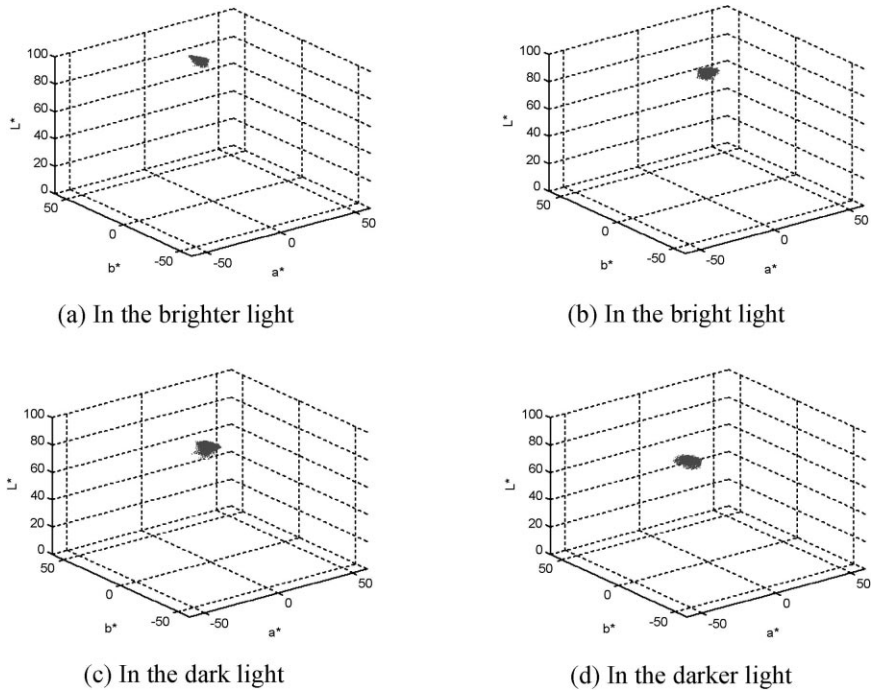
Fig. 4. Skin color distributions under different lighting conditions in *RGB* color space.



(a) In the brighter light (b) In the bright light

(c) In the dark light (d) In the darker light

Fig. 5. Skin color distributions under different lighting conditions in *HSI* color space.



(a) In the brighter light (b) In the bright light

(c) In the dark light (d) In the darker light

Fig. 6. Skin color distributions under different lighting conditions in $L^*a^*b^*$ color space.

1. Skin color distributions under different lighting conditions are highly correlated in *RGB* color space, and have a diagonal translation with the change of lighting conditions.
2. Skin color distributions under different lighting conditions are low correlated in *HSI* color space, and translate along the "*I*" axis with the change of lighting conditions.
3. Skin color distributions under different lighting conditions are low correlated in $L^*a^*b^*$ color space, and translate along the "L^*" axis with the change of lighting conditions.

3.3. Conclusions

According to the above quantitative investigations on skin color distributions of different human hands under different lighting conditions in *RGB*, *HSI*, and $L^*a^*b^*$ color spaces, we obtain the following conclusions:

1. Skin colors of different human hands distribute in a small region in a color space.
2. Skin colors have more difference in intensity than in color.
3. Skin color distributions translate along the "lightness" axis in *HSI* and $L^*a^*b^*$ color spaces with the change of lighting conditions.
4. *HSI* and $L^*a^*b^*$ color spaces are more suitable for hand image segmentation than *RGB* color space.

4. RCE neural network for color segmentation

RCE neural network was designed as a general-purpose, adaptive pattern classification engine [24]. It can resolve pattern recognition problems of lineally and non-lineally separable classes as well as non-separable classes. In this paper, we use it to develop a new color segmentation algorithm.

4.1. Network architecture

The RCE network consists of three layers of neuron cells, with a full set of connections between the first and second layers, and a partial set of connections between the second and third layers, as shown in Fig. 7. In this paper, three cells on the input layer of the RCE network are designed to represent the color values [25], e.g. the *R*, *G* and *B* values, of a pixel in the image. The middle layer cells are called prototype cells. Each prototype cell contains color information about an example of a learned color class that occurred in the training data, and each cell on the output layer corresponds to a different color class presented in the training data set.

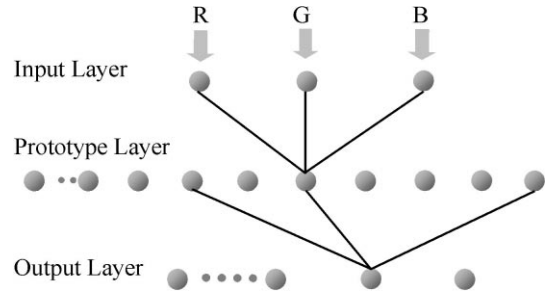


Fig. 7. The architecture of RCE neural network.

The prototype cell of the RCE network is characterized by five elements: class *C*, weight vector $\boldsymbol{\omega}$, cell threshold λ, pattern count *k* and smoothing factor σ. The weight vector $\boldsymbol{\omega}$ represents the set of weighted connections between the prototype cell and each of the input layer cells, and has the same dimension as the input color signal. A prototype cell weight vector thus defines a point in the same color space. The cell threshold λ describes a spherical region of influence around the prototype cell in the color space.

In response to an input color signal $\boldsymbol{X}$, each prototype cell computes a distance between the input color signal and the color prototype vector stored in its weight as follows:

$$d_i = \left[\sum_{j=1}^{3} (\omega_{ij} - x_j)^2 \right]^{1/2}, \tag{5}$$

where ω_{ij} is the weight connecting the *i*th prototype cell to the *j*th input cell, x_j the *j*th color value of the input signal $\boldsymbol{X}$.

A prototype cell will become active to trigger its associated color class *C* if the pattern-to-prototype distance *d* is less than the cell threshold λ. If *d* is greater than or equal to λ, the prototype will not respond to

the input signal. Referring to the output of the ith prototype cell as p_i, this means

$$p_i = \begin{cases} 1 & d_i < \lambda_i, \\ 0 & d_i \geqslant \lambda_i. \end{cases} \tag{6}$$

The pattern count k stores the number of times that the prototype cell has correctly been fired in response to the input signals belonging to its associated class. It is used by the network to approximate the local probability density value for a given color class in the problem of non-separable color distribution.

4.2. Network training

The training procedure of RCE makes use of three mechanisms: prototype cell commitment, prototype threshold modification and prototype pattern count increment. During training, the RCE network allocates the positions of prototype cells and modifies the sizes of their corresponding spherical influence fields by the processes of prototype cell commitment and prototype threshold modification, so as to cover the distribution region for each color class presented in the training data.

Fig. 8 shows the distribution region of skin colors constructed by skin color prototype cells and their spherical influence fields in *RGB* color space. The study on the skin color distribution has indicated that skin colors cluster in a small region in a color space, however, the shape of the skin color distribution region is complicated and irregular. Common color segmentation techniques based on histogram are not effective enough to segment the hand image from the complex and dynamic background due to the difficulty to properly select threshold. In this paper, taking advantage of the adaptive training procedure, numerous skin color prototype cells of the RCE network together with their different spherical influence fields are able to accurately bound the distribution region of skin colors in the color space. So the RCE network is capable to segment various hand images under variable lighting conditions from complex backgrounds after trained properly.

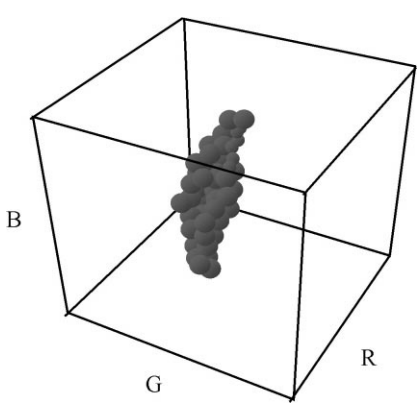


Fig. 8. The distribution region of skin colors in *RGB* color space.

4.3. Network running

During running, the RCE network can utilize either of two response modes to identify input color signals. The first mode is fast response mode, and the second is output probability mode.

4.3.1. Fast response mode

For a input color signal, the RCE network computes the pattern-to-prototype distance d_i for each prototype cell in the network, compares the distance d_i with the threshold λ_i stored with each prototype cell, and identify the prototype cell activity p_i as follows:

If $d_i < \lambda$, then $p_i = 1$.

The active prototype cells trigger the cells on the output layer of the network to which they are connected. In the fast response mode, only a single output layer cell is triggered, and indicates an unambiguous identification of the input signal with that color class.

4.3.2. Output probability mode

If multiple output layer cells are active, or if none is active, the output probability mode is invoked to determine the probabilities that the input signal belongs to the classes represented by the output layer cells. The response of an output cell is an approximation to $P(C|\boldsymbol{X})$, the conditional probability of color class C, given input color signal $\boldsymbol{X}$. To compute the output response, a decaying exponential function is defined for each prototype cell:

$$p_i = \mathrm{e}^{-\sigma_i d_i}, \tag{7}$$

where d_i is the distance between the input signal and the ith prototype cell, σ_i the prototype smoothing factor which controls the decaying rate of function.

If there are N cells on the output layer, and the color class represented by the jth output cell is C_j, then the response of the output cell R_j is given by

$$R_j = \sum_{p_i \in C_j} k_i p_i, \tag{8}$$

and the actual conditional probability $P(C_j|\boldsymbol{X})$ is computed as follows:

$$P(C_j|\boldsymbol{X}) = \frac{R_j}{\sum_{j=1}^{N} R_j}, \tag{9}$$

where k_i is the pattern count for the ith prototype cell.

The output cell with maximum conditional probability indicates an optimal identification of the input signal with that color class.

For hand image segmentation, there is only one cell on the output layer of the RCE network, which represents the skin color class, and the network responds to input color signals in the fast response mode. If an input color signal falls into the distribution region constructed by all the skin color prototype cells together with their influence fields, this input color signal belongs to the skin color class, and the pixel represented by this color signal in the image is identified as skin texture.

The output probability mode of the RCE network can be used for the classification of multiple objects with different colors in the image.

5. Hand segmentation algorithm

The procedure of RCE-based hand image segmentation includes two processes: RCE network training and hand image segmentation. Their algorithms are described as follows, respectively:

1. RCE network train:
   1.1. Select hand areas from color images, and acquire their *RGB* color values. In these areas, representative colors of the hand skin must be included.
   1.2. Convert hand images from *RGB* color space to *HSI* or $L^*a^*b^*$ color space to obtain the training data set of the hand in different color spaces.
   1.3. Train the RCE network using the train set of the hand.
2. Hand image segmentation:
   2.1. Acquire color images to be segmented, and transform their color values from *RGB* color space to *HSI* or $L^*a^*b^*$ color space.
   2.2. Input the color values of the images into the trained RCE network, and the network segments hand images in the fast response mode.

6. Experimental results

This segmentation algorithm has been developed as part of our gesture-based human–service robot interaction system. In the system, the training procedure of the RCE network can be taken either offline or online. Offline training means that the RCE network is trained using the training data saved in a file before segmentation. For online training, parts of the hand section in the video image are selected as the training set of the RCE network directly by the mouse of the computer. The online train set can also be saved and used for offline training.

6.1. Comparison of hand segmentation results in different color spaces

Color images of the hand can be segmented in *RGB*, *HSI* and $L^*a^*b^*$ color spaces in the system. The same section of the hand image shown in Fig. 9(a) is selected as the online training sets of the RCE network in different color spaces. Fig. 9(b)–(d) shows the segmentation results of the hand image with size of 384×288 pixels in *RGB*, *HSI* and $L^*a^*b^*$ color spaces, respectively. The shown segmentation time is 9 s in *RGB* color space, 8 s in *HSI* color space, and 3 s in $L^*a^*b^*$ color space.

The experimental results indicate that segmenting hand images in $L^*a^*b^*$ color space achieves the best effect and consumes the least time. According to the study on skin color distribution as well as the practical segmentation results, we determined that the $L^*a^*b^*$ color space is the most suitable for hand image segmentation.

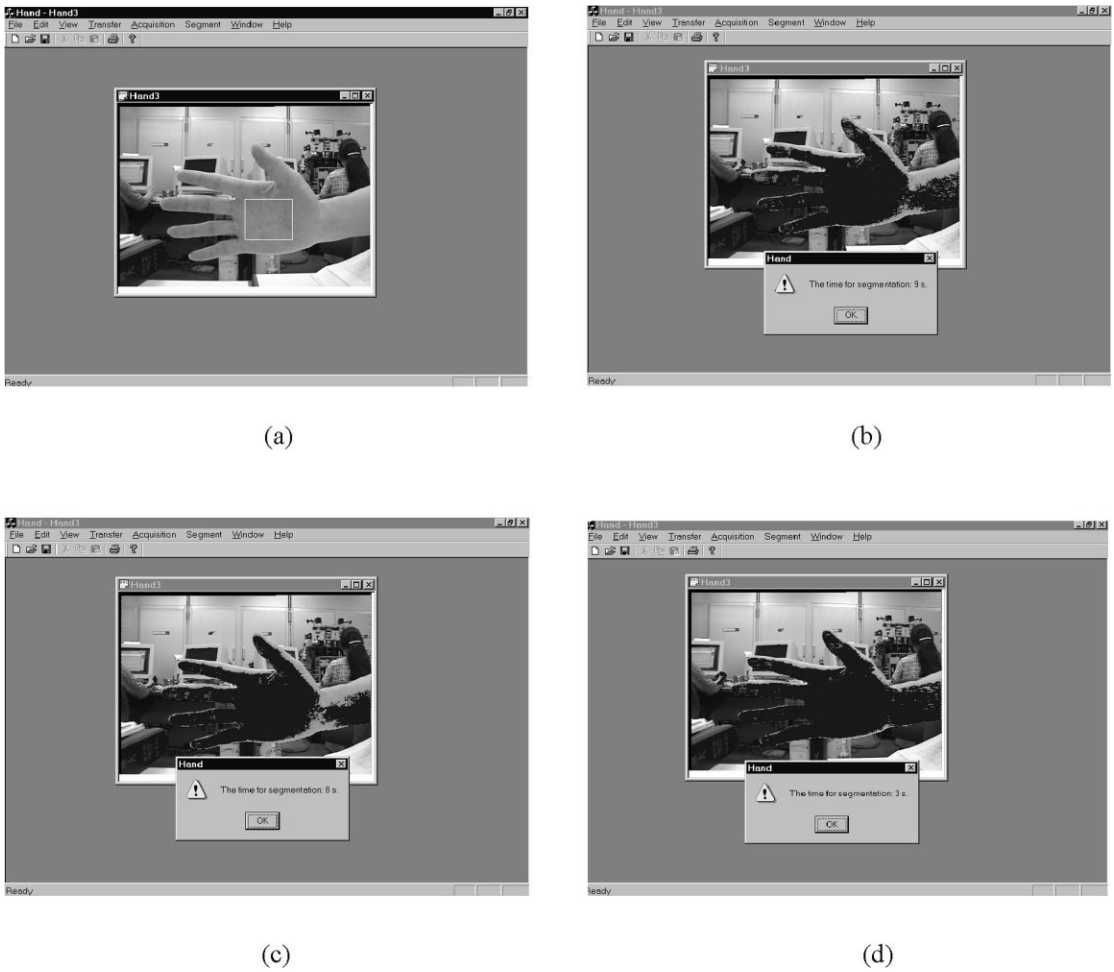


Fig. 9. (a) Online training of the RCE network; (b) hand image segmentation in *RGB* color space; (c) hand image segmentation in *HSI* color space; (d) hand image segmentation in $L^*a^*b^*$ color space.

6.2. Hand region refinement

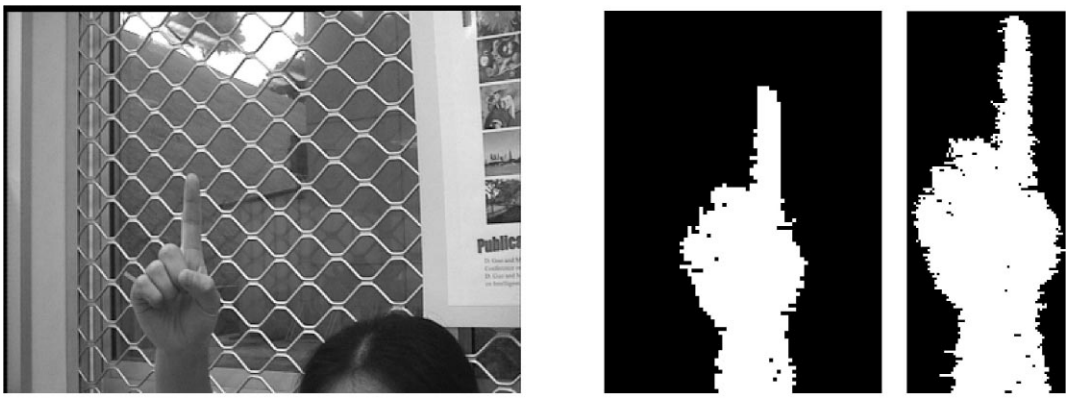
The RCE-based color segmentation algorithm separates all the pixels whose colors are similar to skin colors in the image. There are occasions that other skin-tone objects such as faces are separated, or some non-skin pixels are falsely detected because of the effect of lighting conditions as shown in Fig. 10(a). In such cases, geometric constraints of the hand or some assumptions can be used to further identify the region of the hand. In this paper, we assume the hand is the largest skin-tone object in the image, and use the approach of grouping by connectivity of primitive pixels to refine the hand region. Grouping by connectivity of primitive pixels is a simple image processing technique for pattern recognition, in which the primitive pixels that are detected to belong to the areas of image patterns are grouped into patterns according to the connectivity. Fig. 10(b) shows the result after refinement, in which the region of the hand is segmented perfectly from the complex background.

Fig. 11 shows refined segmentation results of various hand images in a variety of indoor and outdoor

(a) (b)

Fig. 10. (a) Segmented hand region before refinement; (b) segmented hand region after refinement.

Fig. 11. Refined segmentation results of various hand images.



(a) (b) (c)

Fig. 12. (a) The hand image captured by the video camera; (b) hand segmentation for hand tracking; (c) hand segmentation for posture recognition.

environments with different lighting conditions in $L^*a^*b^*$ color space. All the images are taken by the digital camera and with the size of 640×480 pixels.

6.3. Real-time performance

In our human–robot interaction system, live images with the size of 384×288 pixels like Fig. 12(a) are captured through two CCD video cameras (EVID31, SONY). The segmentation algorithm described above is used in two phases: hand tracking phase and posture recognition phase. At the beginning, we have to segment the whole image to locate the hand, because we do not have any information on the position of the hand. After the initial search, we do not need to segment the whole image, but a smaller region surrounding the hand, since we can assume continuity of the position of the hand during tracking. At the tracking phase, the hand is segmented from a low resolution sampling of the image, and can be tracked reliably at 3–5 Hz on a normal 450 MHz PC. The system also detects the motion features of the hand such as pauses during tracking phase. Once a pause is confirmed, the system stops tracking, then crops a high-resolution image tightly around the hand and performs a more accurate segmentation based on the same techniques. The segmented hand image is used to extract the topological features of the hand and the classification of various hand postures is based on the analysis of these features. The time spent on the segmentation of the high-resolution image is less than 1 s, and the whole recognition phase can be accomplished within 1.5 s. After the posture recognition phase is finished, the system continues to track the hand until another pause is detected.

Fig. 12(c) and (d) show the examples of segmentation for hand tracking and posture recognition using the hand segmentation approach presented above. Their quality of segmentation is better obviously, compared with Fig. 2 in [14], which shows the examples of segmentation for tracking and pose recognition using the approach proposed by Kjeldsen and Kender. Preliminary trials were conducted on our human–robot interaction system with users of different age, gender and skin color. The system successfully recognized each of the gestures with the accuracy of over 95% after the RCE network was trained properly. The recognition accuracy may decrease in the case that the user and lighting condition changed too much, due to the previous training of the RCE network becoming insufficient. However, this problem can be solved easily by selecting parts of undetected hand sections as the training data using the mouse, and incrementally performing the online training. There is no need to re-represent the entire training set to the network.

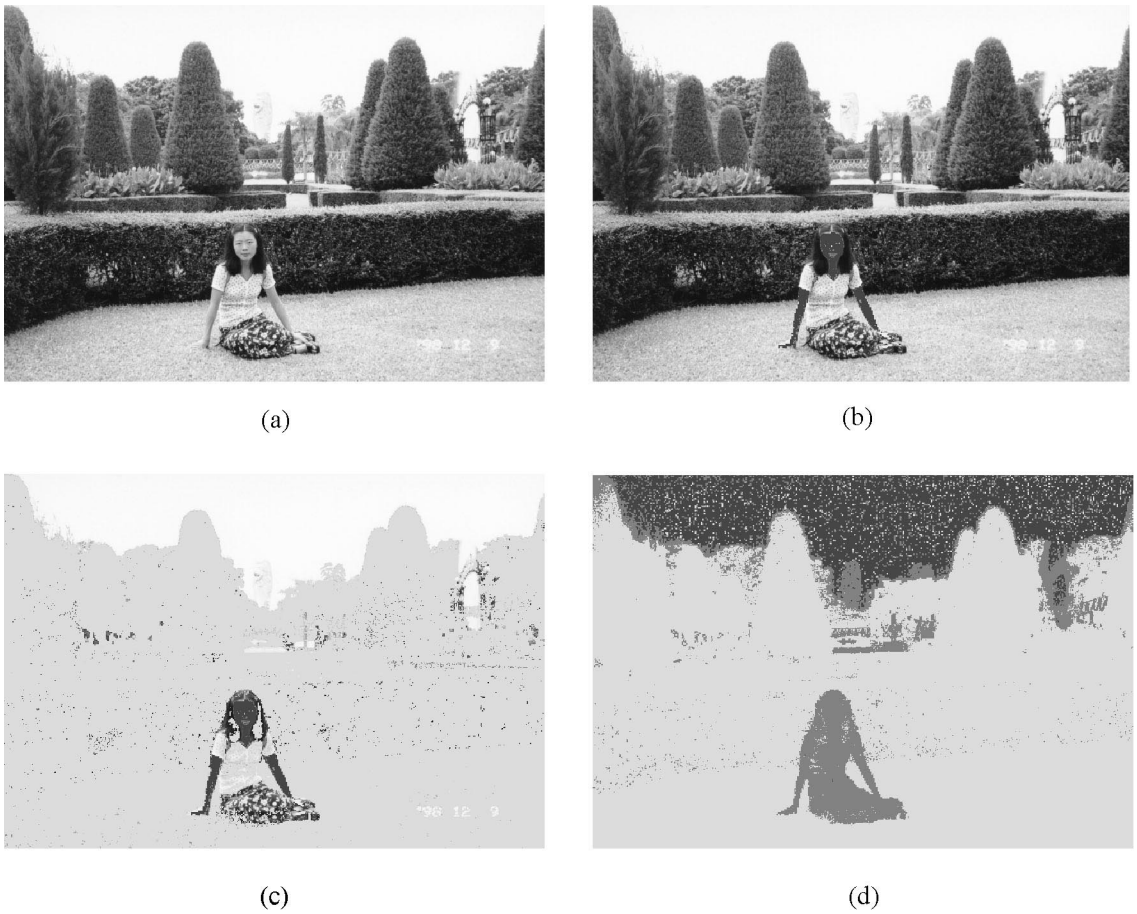


Fig. 13. (a) A color photo; (b) segmentation of the human skin; (c) segmentation of the skin and the trees; (d) segmentation of the whole image.

7. General color image segmentation

The RCE neural network is a general-purpose and adaptive pattern classification engine. Our system thus can also do segmentation for arbitrary color images. The method is similar to that described above, the only difference being that the pattern class of each selected color object is specified at the same time when the training set of that color object is constructed. The different objects with different colors can be defined as the same pattern class. After training, the RCE network can segment one or more interesting objects in the image in the fast response mode, or segment the whole image in both the fast response mode and the output probability mode.

Fig. 13(a) is a color photo of 728×499 pixels, and Fig. 13(b) shows the segmentation result of the human skin in $L^*a^*b^*$ color space. Fig. 13(c) shows the segmentation result of the human skin and the trees, and Fig. 13(d) is the segmentation result of the whole image, in which the hair, the skin and the clothes of the person are defined as the same pattern class.

Fig. 14(a) shows a color picture of 512×483 pixels, and the segmentation results of the trees, the road and the sky as well as the whole image in $L^*a^*b^*$ color space are shown in Fig. 14(b)–(d), respectively.

Fig. 14. (a) A color picture; (b) segmentation of the trees; (c) segmentation of the road and the sky; (d) segmentation of the whole image.

8. Conclusion

Hand image segmentation is the first step in hand gesture recognition systems. In this paper, a novel method for hand image segmentation under the context of gesture-based human–service robot interaction is presented. This method uses color information of the human skin and is based on RCE neural network.

The study on skin color distributions in *RGB*, *HSI* and $L^*a^*b^*$ color spaces indicates that human skin colors cluster in a small region in a color space and have more difference in intensity than in color. During training procedure, the RCE neural network accurately characterizes the arbitrarily complex skin color distribution region in the color space with numerous skin color prototype cells together with their different

spherical influence fields, and during running procedure, the RCE network identifies all the skin-tone pixels in the image in the fast response mode. Resorting to the assumption that the hand is the largest skin-tone object in the image, we use the technique of grouping by connectivity of primitive pixels to refine the hand region in this paper. Experimental results demonstrate that this segmentation method can segment various hand images efficiently from all kinds of complex backgrounds in real time, and $L^*a^*b^*$ color space is the most suitable for hand image segmentation. This RCE-based color segmentation algorithm can also be applied for general segmentation of arbitrary color images.

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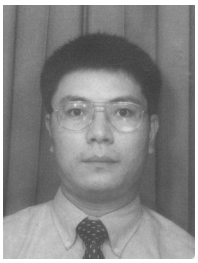

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